北京理工大學

本科生毕业设计（论文）

挖掘关联规则的市场篮子分析方法

Using Associative Mining Rules for Market Basket Analysis

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挖掘关联规则的市场篮子分析方法

摘 要

数据离不开日常生活，是非常宝贵的资源。 一个大大小小的组织都可能充斥着各种数据，以及一家商店的每笔销售交易所涉及的数据，例如购买的商品数据、购买时间、购买的商品数量、商品价格和消费者数据。 由于每笔交易的数据量都很大，而且数据增长很快，所以数据集被闲置的情况并不少见。 因此，我们可以从数据集中挖掘信息，使其成为有用的信息来支持决策或协助确定营销策略，因为零售业在东帝汶是一个快速发展的行业，与消费者的日常生活有着非常密切的关系，在东帝汶，许多线上零售商店使用不同的策略来改进他们的营销策略，使用关联规则的购物篮分析越来越成为主流策略。

挖掘关联规则是用来购物篮分析（MBA）的挖掘与分析数据的技术之一。MBA用于分析零售环境中的消费者行为并发现一起购买的产品的组合。 例如，买牛奶的顾客也买了面包，或者买鸡的人也买了米饭。 MBA 是一组突出购买趋势的统计亲和力计算，可帮助企业高管更好地了解和服务客户。MBA 最初扫描交易以寻找最典型的产品配对。本文分析中，使用了位于东帝汶首都帝力的一家零售店 X 的交易数据，涉及 2022 年 1 月至 4 月的交易，将其分为 4 个时期，涉及 18818 条交易记录和 4200 件售出的独特商品，挖掘每个时期生成的关联规则，将所有的规则缩短，利用关联规则的整体可变性，找出最重要的规则，帮助营销部门制定更好的营销策略。

关键词：购物篮分析（MBA），关联挖掘规则，营销策略。

Using Associative Mining Rules for Market Basket Analysis

Abstract

Data cannot be separated from everyday life and is a very valuable resource. An organization both large and small can be flooded with various kinds of data as well as the data involved in every sales transaction at one store, such as item data purchased, time purchases, number of items purchased, item prices, and consumer data. Because of the large amount of data in each transaction and the rapid growth of data, it is not uncommon for data sets to be left alone. Therefore, we can mine information from the data set and make it useful information to support decisions or assist in determining marketing strategies as retails is a rapid growing industry in Timor-Leste, many retail store has use different strategy to improve their marketing strategy but none of any retails store used association rules for their market basket analysis.

Associative mining rules is one of the techniques we can use to analyze data through **Market basket analysis**. MBA or market basket analysis is a data mining technique for analyzing consumer behavior in any retail environment[1]. It's simple terms basically is to analyze the combination of products which have been bought together. For Example, customers who bought milk also bought bread or people who bought chicken also bought rice. MBA, a set of statistical affinity calculations that highlight purchase trends, helps business executives better understand and serve their client. MBA initially scans transactions for the most typical product pairings. In this analysis is used a transaction data of a supermarket X located in Dili, the capital of Timor-Leste involving transaction from January to April of 2022 that we divided to 4 period involving 18818 Transaction record and 4200 unique items sold, after all the association rule generated from each period, all the rules will be shorted using overall variability of association rules to find most important rules to help the marketing department make a better marketing strategy.

Keywords: Market Basket Analysis (MBA), Associative Mining Rules, Marketing Strategy.

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Chapter 1: Introduction

1.1. Introduction of Associations Rules Analysis

Association Analysis is very useful for discovering interesting relationship hidden in large datasets. The relationship between items in one column or basket can have a very useful insight, as an example of market basket analysis of retail stores, the transaction record of customers and the items they bought in their basket can be used for analysis to help the marketing department make a better strategy to up sale their products.

Table 1-1 An example of market basket transaction

TID	ITEMS
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

The uncovered relationship can be represented in the form of association rules or sets of frequent items, for example, the following rule can be extracted the data set shown in table 1[1].

{Diapers} → {Beer}.

The rule tells that a strong relationship exists between the sale of diapers and beer because many customers who buy diapers also buy beer. The retailer company can use this type of rules to help them identify new opportunities for cross-selling their product. Beside market basket data, association analysis is also applicable to other application domains such as bioinformatics, medical diagnosis, Web mining, and scientific data analysis. Earth science data analysis also used association rules for analyze association pattern that may reveal interesting connections among the

ocean, land and atmospheric process. This kind of information can help Earth system interact with each other. Even though the technique presented here are generally applicable for a wider variety of datasets, for the purpose of this thesis will focus mainly on market basket analysis for retail store[2]. Association mining rule for Market basket analysis is a technique commonly used in retail and e-commerce to analyze customer purchasing behavior and identify associations between products. It aims to discover which items are frequently bought together, allowing businesses to make strategic decisions regarding product placement, promotions, and cross-selling.

Association mining rules play a crucial role in market basket analysis as they uncover meaningful relationships and patterns within transactional data. By identifying which items are often purchased together, businesses can gain insights into customer preferences, optimize inventory management, and customize marketing strategies.

The process of association mining for market basket analysis typically involves the following steps:

- **Data Collection**

  Transactional data is collected, which consists of records of customer purchases. Each transaction represents a basket of items bought by a customer.
- **Preprocessing**

  The transactional data is prepared for analysis by removing any irrelevant information, handling missing values, and encoding the data in a suitable format for association rule mining algorithms.
- **Item Frequency**

  The frequency of each item is calculated, indicating how often it appears in the transactions. This information helps identify popular items and potential associations.
- **Support Calculation**

  The support of item sets is computed, representing the proportion of transactions in which a particular itemset occurs. Item sets with high support are considered significant and likely to form association rules.

- **Association Rule Generation**

  Using algorithms such as Apriori or FP-growth, association rules are generated based on the support and confidence thresholds set by the analyst. These rules express the relationships between different items and provide insights into purchasing patterns.
- **Rule Evaluation**

  The generated rules are evaluated based on metrics such as support, confidence, and lift. Support measures the frequency of the itemset occurrence, confidence measures the conditional probability of the consequent item given the antecedent, and indicates lift the strength of the association between the items.
- **Interpretation and Action**

  The generated association rules are interpreted to gain insights into customer behavior. Businesses can then use this information to improve product placement, create targeted marketing campaigns, suggest related products, or optimize pricing strategies.

By leveraging association mining rules for market basket analysis, businesses can enhance their understanding of customer preferences, increase sales, and optimize their operations. It enables them to make data-driven decisions that cater to customer needs, improve customer satisfaction, and drive business growth. Overall, association mining rules provide a valuable technique for uncovering meaningful associations in market basket data, helping businesses gain insights into customer behavior and make informed decisions to improve their overall performance, for more details about how the association rules work and the algorithm will discuss in chapter 3.

1.2. Introduction of Association Rules for Market Basket Analysis

In market basket analysis, associations and patterns between the products that customers purchase is found using the potent technique of association rule mining. Businesses can use this information to determine links between regularly purchased products, enabling them to plan their product placement, cross-selling efforts, and targeted marketing campaigns. Finding interesting associations or links between items in a transactional dataset is the aim of association rule mining.

These relationships are expressed as "if-then" rules, where the presence of one thing in a transaction indicates that it is likely that another item will also be present. For instance, association rules in a grocery store environment can provide information like "if a customer buys bread and milk, then there is a high likelihood that they will also purchase eggs." Retailers may find these associations useful for enhancing their product selection, store design, and marketing initiatives[3]. Typically, association rule mining contains two crucial steps:

Frequent Itemset Generation: In this step, groups of items that frequently appear together in transactions are identified. Frequent item sets are these collections of items. The percentage of transactions in which an item appears is indicated by its support. The transactional dataset can be used to extract frequent item sets by using techniques like Apriori or FP-Growth. Once the frequent item sets have been acquired, association rules are constructed by looking at the connections between the items in the item sets. These regulations are founded on variables like support, confidence, lift, or leverage. Lift measures the strength of the association between items, support measures the frequency of occurrence of an item set, confidence indicates the likelihood of the consequent item given the antecedent item and leverage measures the discrepancy between the observed frequency and the expected frequency of the items occurring together. Based on these indicators' user-defined thresholds, association rules are assessed.

The rules that emerge are then categorized and examined according to how intriguing and potentially influential they are for business[4].

In conclusion, by identifying significant relationships between items, association rule mining in market basket analysis offers insights into consumer purchasing behavior. Businesses can maximize their cross-selling, targeted marketing, and product placement strategies by utilizing these relationships, which will ultimately improve consumer happiness and increase sales income.

By examining common patterns and utilizing support and confidence factors, association rules are created. According to Tan et al, support determines how frequently a rule applies to a particular data set. From the perspective of business, a low support rule is uninteresting, and occasionally it

will be removed. While confidence determines how frequently items in Y appear in transactions that contain X, the higher the confidence, the more likely it is for Y to appear in transactions that contain X.

$$\text{Support, S } (X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N} \tag{1-1}$$

$$\text{Confidence, C } (X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} \tag{1-2}$$

Lift ratio is one of the parameters that is also used to determine important rules in AR. Lift ratio is a parameter that is used to see whether the rules gained from AR are strong or not. The lift ratio measures the possibility of X and Y occurring together divided by the possibility of X and Y occurring if they are independent events. The formula for the ratio is as follows:

$$\text{Lift Ratio, LR} = \frac{Support(X \cup Y)}{Support(X)\,.\ Support\ (Y)} \tag{1-3}$$

Association rules are useful in data mining for analyzing and predicting consumer behavior and play an important role in the analysis of shopping basket data, product clustering, catalogue design, and store plan layout. In addition to supporting product analysis, association rule mining is also used to identify operational issues in buildings' heating, ventilation, and air conditioning (HVAC). In the discipline of strategic management, the association rule is also used to identify successful management approaches. The association rule technique is applied with a number of algorithms. Apriori, Eclat, and FP-growth Algorithms are just a few of the algorithms that have been employed by researchers, according to Dhanalakahmi and Porkodi's research. The association rule algorithm was created using a novel technique by Sandhu et al. based on the assumptions of profit and quantity.[5]

This analysis generated a set of association rules from the database using the apriori technique. For the Boolean association rule, the Apriori algorithm is useful for identifying frequent items. The apriori algorithm's name derives from the fact that it frequently employs prior knowledge from item mining searches. The number of products, the volume of transactions, and the size of the market basket all affect the original Apriori algorithm. The study suggested a practical method for effective data mining from highly useful association rules based on weighting factors and utilities.

In their study, Kaur and Kang outlined how several algorithms used in existing AR only functioned on static data and were unable to catch real-time changes in data[6]. The method suggested in this study not only processed static data but also recorded dataset changes. application of MBA in a setting with several stores. When used in supermarkets with several stores of varied sizes, frequent product changes, and longer-period datasets, the purposed method was found to be more computationally efficient than the old method.

Identifying and predicting consumer behavior based on past client buying habits is one of the objectives of the MBA. MBA is used in a variety of sectors in addition to supermarkets. There are to be compared with other methods to propose a product network analysis, a network-based analysis to analyze a network-levelled relation between products. Artificial Neural Network (ANN) back propagation is another approach that the MBA uses in combination with other methods to estimate product inventory. A hospital setting might benefit from an MBA. By helping hotels identify the most desirable ancillary goods and services (beyond the room type) to provide to new and returning hotel guests, market basket analysis could boost revenue.

1.3. Applications

Association rules give retailers useful information for market basket research, allowing them to better understand consumer behavior, improve product offerings, focus marketing efforts, and boost operational effectiveness. Retailers can use these rules to make data-driven decisions that result in increased sales, client satisfaction, and expanded business, here are some applications that association rules can provide:

- Uncover Purchasing Patterns
- Cross-Selling and Upselling Opportunities
- Personalized Marketing
- Product Placement and Store Layout Optimization
- Inventory Management and Stock Optimization
- Market Basket Analysis for Pricing Strategies
- New Product Development

1.4. Motivation

The motivation behind conducting market basket for retail store using Association rules is come myself as I am also currently starting my small retail business in textile and apparel sector, however in thesis I am conducting my analysis in the dataset of a supermarket X located in Timor-Leste, beside my very personal motivation the retail industry in Timor-Leste also having a rapid growth since the last decade, therefore I hope the result of this Thesis can good paper to help the retailers mining their day to day transaction records to a valuable resources and Perform the market basket analysis to help increase their sales and manage their supermarket.

1.5. Objective

The goal of market basket analysis is to find relationships and patterns among the goods that customers have purchased in a transactional dataset. It strives to offer insightful information about consumer behavior, item co-occurrence, and shopping preferences. The following are the main goals of market basket analysis:

- **Finding Item Associations**

Market Basket Research looks for products that customers typically purchase together. Retailers can learn about the connections between various products as well as consumer preferences and buying habits by identifying associations.

- **Producing Association Rules**

  The goal of market basket analysis is to provide association rules that give retailers useful information. These rules often take the form of "if-then" statements and express relationships between things. By way of illustration, "If a customer buys item A and item B, then they are likely to purchase item C." Making intelligent choices for product placement, cross-selling, and targeted marketing efforts is made easier by these association rules[7]

The main objective of the overall variability of association rules in market basket analysis for a retail store is to shorten out the most important rules that frequently appear in each period and also quantify the variety and range of links between items in customer transactions. It offers perceptions of consumer behavior, directs decisions about product assortment, supports focused marketing initiatives, and enhances shop design and product placement tactics. Retailers may fully comprehend client preferences and adjust their operations by taking total variability into account.

1.6. Problem Definition

Most retailers in Timor-Leste don’t mine or organize their everyday transaction data or even use these for proper analysis to help the marketing department make better decisions. With this type of problem, the use of association rules through market basket analysis can help these retailers perform a proper analysis with their everyday data for a much wider benefit. The marketing

strategy of retail establishments that do not use market basket research may be fraught with difficulties and chances lost. These are a few issues that they could run into:

- **Ineffective Cross-selling**

Retailers may find it difficult to discover organic relationships between products in the absence of market basket data. Due to their ignorance of the products that clients frequently purchase together, they miss out on cross-selling opportunities. Due to this, they lose out on prospective sales and see a decline in the efficiency of their marketing initiatives.

- **Ineffective Product Placement**

Retailers may find it difficult to optimize their product placement throughout the store without the knowledge gained from market basket analysis. They might not know which goods are usually bought together, resulting in less-than-ideal store layouts that inconvenience customers and reduce sales. Lack of Personalization: If retailers do not make use of market basket analysis, they may find it difficult to personalize their marketing campaigns. They are unable to provide targeted recommendations, promotions, or personalized offers that can improve customer happiness and boost sales since they do not comprehend specific client preferences and purchasing behaviors.

- **Missed Inventory Optimization**

Retailers may struggle to properly optimize their inventory management without a market basket analysis. They could not have proper knowledge of the relationships between items and demand trends, which could lead to ineffective stock replenishment, excess inventory, or stockouts, all of which have a detrimental effect on customer satisfaction and profitability. Limited Customer Insights: Market basket analysis offers useful information about customer preferences and behavior. The ability of retailers to make data-driven decisions and customize their goods to customers' needs is limited by a lack of awareness of purchase patterns, customer segmentation, and customer preferences.

- **Ineffective Promotions and Campaigns**

Without a market basket study, retailers may find it difficult to create promotions and marketing strategies that will be successful. They could be unable to choose the best products to bundle or suggest, leading to less effective marketing tactics and missed chances to increase sales and consumer engagement. Overall, a retail store's marketing may suffer from missing cross-selling possibilities, bad product placement, a dearth of personalization, poor inventory management, limited consumer insights, and ineffective promotions if market basket strategy research is not conducted. Adopting market basket analysis can help retailers overcome these obstacles and improve consumer happiness while increasing their bottom line.

Chapter 2: Literature Review

2.1. Introduction

In this section, will be discuss some of the past studies related to Association rule for market basket analysis. By synthesizing and analyzing the existing literature, this review aims to identify the advancements made in association analysis, the challenges encountered, and the potential avenues for future research. It will serve as a valuable resource for researchers, practitioners, and organizations seeking to leverage association analysis techniques to gain deeper insights into customer behavior and optimize their business strategies. Overall, association analysis has emerged as a critical tool for understanding item associations and customer purchasing behavior. This literature review aims to consolidate the knowledge in this field, providing a foundation for further research and practical application.

2.2. Related Papers

2.2.1. Fast Algorithms

Rakesh Agrawal et al. first studied the issue of mining association rules over basket data. An illustration of such a rule would be that 98% of customers who buy tires and car accessories also get their vehicles serviced. Finding all of these rules is useful for applications like attached mailing and cross-marketing[8], Catalogue design, add-on sales, retail architecture, and customer segmentation based on purchasing trends are other applications. These applications use incredibly huge databases. Therefore, having quick algorithms is essential for this activity.

- **Applications and Implications**

The authors go over a few of the association rule mining's potential applications, such as market basket analysis, cross-marketing, catalogue creation, and inventory management. They emphasize the importance of association rules in supplying information about consumer behavior and guiding company decisions.

2.2.2. Multi-level Apriori Algorithm

The paper by Rakesh Agrawal, Tomasz Imieliński, and Arun Swami introduces the concept of mining association rules between sets of items. It presents the Multi-level Apriori algorithm, which enables the discovery of set-based association rules at different levels of item granularity. The paper discusses the motivation, algorithm steps, experimental evaluation, and applications of set-based association rule mining [9].

- **Applications and Implications**

The authors discuss the potential applications of set-based association rule mining, including market basket analysis, customer segmentation, and collaborative filtering. They highlight the value of discovering relationships between sets of items in understanding complex patterns and providing more comprehensive insights for decision-making.

2.2.3. Mining Maximal Frequent Item Sets

The concept of mining maximal frequent item sets is introduced in the publication by M.J. Zaki and K. Gouda, who also suggest the MFIM algorithm for effective mining. It covers the rationale, algorithm phases, approaches for improving efficiency, experimental analysis, and applications of mining most frequent item sets. The research makes frequent itemset mining more effective and interpretable, allowing for more rapid examination of massive datasets and the generation of insights that are clearer and more insightful [10].

- **Applications and Implications**

The authors discuss the implications of mining maximal frequent itemsets in various applications, including market basket analysis, web usage mining, and bioinformatics. They highlight the advantages of maximal frequent itemsets in providing more concise and meaningful results, enabling faster decision-making and more focused analysis.

2.3. Retailers Industry

The retail industry in Timor-Leste has grown rapidly since the last decade. The retail sector is the most prevalent form of business in Timor-Leste. In more sophisticated types of activity, in 2021,

value added in wholesale, retail trade, restaurants, and hotels for Timor-Leste was -3.8%. Though Timor-Leste's value added in wholesale, retail trade, restaurants, and hotels fluctuated substantially in recent years, it tended to decrease through the 2002–2021 period, ending at -3.8% in 2021. For both traditional and online retailers, forecasting future demand is crucial. It helps keep the right amount of inventory on hand, which by itself raises sales and improves customer satisfaction. Retailers benefit from having a deeper understanding of customer behavior[4]. Forecasting future sales volumes using time series analysis over aggregated customer sales data is a frequent and well-researched approach to demand prediction. Market basket analysis (MBA) is another viewpoint. Although it is currently far less researched in the literature, it has many potential advantages for merchants. It helps them manage their supply chain better and is advantageous from a marketing perspective. To stay competitive among retailers, marketing strategy is one of the key metrics to improve their sales. In order to make a better marketing strategy, retailers have to understand their consumers' behavior through their everyday transaction data using a common technique called market basket analysis.

- **Overview of the Retail Industry in Timor-Leste**

  Traditional Markets: Timor-Leste's traditional markets are important to the country's retail industry. Local sellers sell a variety of things at these markets, which are frequently found in rural and urban regions. These goods include fresh produce, food, home goods, clothing, and handicrafts. Locals go to traditional markets frequently for their everyday purchasing needs, and they act as significant economic hubs.

- **Small Shops and Convenience Stores**

  Timor-Leste is filled with small shops and convenience stores, particularly in urban areas. These businesses provide a variety of goods, including food, drinks, toiletries, and other household items, to meet the immediate requirements of their patrons. Residents who want to shop near their houses might benefit from these stores' convenience.

- **Supermarkets and hypermarkets**

In Timor-Leste's urban centers, modern retail forms, including supermarkets and hypermarkets, have begun to appear. Groceries, household goods, technology, apparel, and other items are all available in these larger-scale businesses. They offer a one-stop shopping experience and frequently stock products from both national and international companies.

- **Shopping Centers**

  Timor-Leste's urban regions have begun to see the development of shopping centers, which offer a variety of services, restaurants, and retail establishments. These shopping malls appeal to customers looking for a wide variety of goods and recreational activities because they provide a more contemporary and organized shopping experience.

- **E-commerce and Online Shopping**

  E-commerce and online shopping platforms have become more popular in Timor-Leste as a result of the expansion of internet access and smartphone use. Customers can buy conveniently from home with online shops and delivery services, giving them access to a wider selection of goods.

It's significant to remember that Timor-Leste's retail sector is still modest when compared to more advanced nations. The development and growth of the sector may be hampered by issues including inadequate infrastructure, logistics in the supply chain, and financial accessibility. The retail sector is, however, gradually transforming as a result of urbanization, economic growth, and shifting consumer tastes. Local business owners, as well as regional and international investors, have the chance to support the growth of Timor-Leste's retail industry. The retail sector is anticipated to grow more as the nation develops, providing consumers in Timor-Leste with better shopping experiences and a greater choice of goods.

Chapter 3: Methodology

3.1. Introduction

In this chapter will be describe a details analysis methodology conducting for this thesis and will describe a brief understanding about Association rules, apriori algorithm, F-P growth and overall variability of association rules, there is also step that used in this analysis to generate association rules, the steps are as follows.

1. the List items by customer IDs.
2. Create an item matrix.
3. Get the support value using the Apriori algorithm.
4. Create a data frame with product support, confidence, and lift values to generate. association rules for each period.

3.2. Associations Rules

Here are some common algorithms that are used to generate frequently bought item sets and association rules:

- Apriori Algorithm
- FP-growth Algorithm

3.2.1. Apriori Algorithm

Agrawal et al.'s 1993 Apriori method was one of the first and best algorithms for mining all frequent item sets and Association Rule Mining[11]. Apriori algorithmic programs are designed to make several runs across the database. The anti-monotonic feature of set theory, which states that every set of the frequent itemset is extra frequent, is the foundation of the Apriori (level-wise method)[12]. Apriori may also function as a candidate generation algorithmic programmed and present problem in a very level-wise manner. To efficiently count candidate item sets, a tree structure and breadth first search are used.

The Apriori property employs the following two steps:

- Join step: - Ck is generated by combining lk–1 with itself.
- Prune Step: - Any (k – 1) item set that's not frequent cannot be a set of a frequent k item set.

3.2.2. Apriori Algorithm Pseudocode

Apriori Algorithm Pseudocode (3-1)

```
Procedure Apriori (T, mSupport) {//T is the database and mSupport is that the minimum Support}
L1 = {frequent items};
For (k= 2; Lk-1 ! =∅; k++) {Ck= candidates generated from Lk-1
For each transaction t in database do {Increment the count of all candidates in Ck that
are contained in t Lk = candidates in Ck with mSupport}
//end is for every statement}
//end is for
ReturnUkLk;
}
```

3.2.3. Benefits of Apriori Algorithm

- Large itemset could be utilized.
- It is extremely convenient and simple for implementation

3.2.4. Drawbacks of Apriori Algorithm

- The Apriori algorithmic program takes longer time for candidate generation technique.

- The Apriori algorithmic program needs many scans of the database.
- Many trivial rules are derived and it will be hard to extract the most interesting rules.
- Rules can be inexplicable and fine grained.
- Redundant rules are generated.

Beside the advantage and the disadvantage that the Apriori algorithm have, however there is a method that can be used to improve the efficiencies of the apriori algorithm, the method is as follows:

1. Hash-based itemset counting: A k-itemset whose corresponding hashing bucket count is below the threshold cannot be frequent.
2. Transaction reduction: A transaction that does not contain any frequent k-itemset is useless in subsequent scans.
3. Partitioning: Any itemset that is potentially frequent in DB must be frequent in at least one of the partitions of DB.
4. Sampling: mining on a subset of given data, lower support threshold + a method to determine the completeness.
5. Dynamic itemset counting: add new candidate item sets only when all of their subsets are estimated to be frequent.

3.2.5. FP Growth Algorithm

The Apriori algorithmic program is based on the anti-monotonic property. The two main issues are repeated database scans and high execution times. There is a need for a compact data structure for mining frequent items. The FP growth algorithmic program is an efficient algorithm for producing frequent item sets without generating candidate item sets. It adopts a divide and conquer strategy, and it needs two database scans to seek out the support count. It can mine the items by using lift, leverage, and conviction by specifying a minimum threshold[13].

3.2.6. Generating FP-Trees Pseudocode

The Algorithm work as follows (3-2)

1. Scan the transaction database once, as with the Apriori algorithmic program, to seek out all the frequent items and their support.

2. Sort the frequent items in descending order of their support.

3. Initially, begin making the FP-tree with a root "null".

4. Get the primary transaction from the transaction database. Take away all non-frequent items and list the remaining items in line with the order among the sorted frequent items.

5. Use the transaction to construct the primary branch of the tree, with each node corresponding to a frequent item and showing that item’s frequency. That’s one for the primary transaction.

6. Get the next transaction from the transaction database. Take away all non-frequent items and list the remaining items in line with the order among the sorted frequent items.

7. Insert the transaction within the tree using any common prefix that may appear. Increase the item counts.
8. Continue with step 6 until all transactions in database are processed.

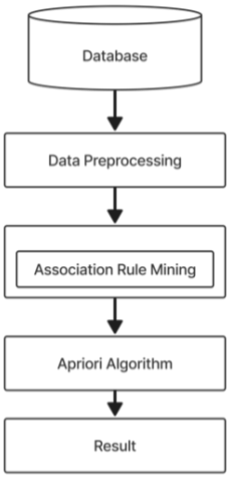

Figure 3-1 Association rule mining in mining

Both the Apriori and FP Growth algorithms are used to mine the frequent patterns from the database. Both algorithms use some technique to discover frequent patterns. The Apriori algorithm works well with large databases, but the FP Growth algorithm works badly with large databases. The comparisons between both algorithms based on technique, memory utilization, number of scans, and time consumed are given below:

- **Technique**

The Apriori algorithm uses the Apriori property and the join, pure property for mining frequent patterns. The FP Growth algorithm constructs conditional pattern free and conditional pattern base from the database, which satisfy the minimum support.

- **Search Type**

Apriori uses the breadth-first search method, and FP Growth uses the divide-and-conquer method.

- **Memory Utilization**

The Apriori algorithm requires large memory spaces as it deals with a large number of candidates itemsets. The FP Growth algorithm requires less memory due to its compact structure; it discovers frequent item sets without candidate item set generation[13].

- **Number of Scans**

The Apriori algorithm performs multiple scans for generating the candidate set. The FP Growth algorithm scans the database only twice.

- **Time**

In the Apriori algorithm, execution time is more wasted in producing candidates every time. FP Growth's execution time is less when compared to Apriori[14].

3.3. Overall Variability of Association Rules (OCVR)

3.3.1. How an Overall Variability of Association Rules Works

Due to unpredictable consumer purchasing patterns, the values of support, confidence, and lift ratio in the results of AR frequently change over time. The variability index of parameter changes in AR is known as OCVR. This variability indicator is calculated using the standard deviation concept from statistical analysis. The formula below is used to determine the variability index (CV).

$$\mathrm{CV} = \frac{S}{X} \qquad \text{Formula (3-1)}$$

S stands for standard deviation, and X stands for average. Index Variability Lift (CVL) and Index Variability Confidence (CVC) are both calculated using this variability index. So, using CVL and CVC, the analysis of the Overall Variability of Association Rule (OCVR) can be expressed as follows:

$$\mathrm{OCVR} = \frac{CVL+CVC}{2} \qquad \text{Formula (3-2)}$$

The OCVR analysis's findings demonstrate how much AR varies from one time to the next. These findings can be taken into account while deciding which rules are crucial in MBA. Additionally, the rule with a high OCVR value, which denotes significant shifts in consumer purchasing patterns over time, may be expanded. The decision-maker can create its advertising strategies based on the results.

3.3.2. How the OCVR Algorithm Works

Overall variability of association rule Pseudo code (3-3)

1. Find a variability index with for confidence and lift ratio with this formula:
   CV (variability index)
   CV = S/X
   S (Standard deviation)
   X (Average)

2. CVC = S/X (to find the variability index for confidence)
3. CVL = S/X (to find the variability index for lift ratio)
4. Use the value of CVC and CVL to find the overall variability of association rule with this formula: OCVR = CVL + CVL / 2

3.4. Analysis Methodology

3.4.1. Introduction

In this section will be describe briefly about Market basket analysis for a retail store or a supermarket that located in Timor-Leste using Association rules, the algorithm that is used is Apriori algorithm and Python as a programing language to run the algorithm, the flowchart of this analysis is shows in bellow:

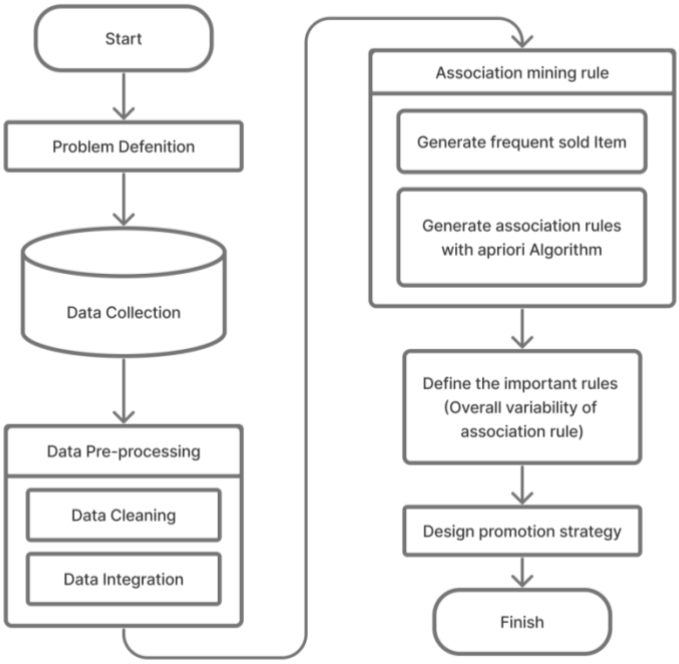

Figure 3-2 Association rule for market basket analysis methodology flowchart

Association mining rule for Market basket analysis is a technique commonly used in retail and e-commerce to analyze customer purchasing behavior and identify associations between products. It aims to discover which items are frequently bought together, allowing businesses to make strategic decisions regarding product placement, promotions, and cross-selling.

Association mining rules play a crucial role in market basket analysis as they uncover meaningful relationships and patterns within transactional data. By identifying which items are often purchased together, businesses can gain insights into customer preferences, optimize inventory management, and customize marketing strategies.

The process of association mining for market basket analysis typically involves the following steps:

- Data Collection
- Preprocessing (Data Cleaning and Data Integration)
- Support Calculation

- Rule Generation
- Interpretation and Action

3.4.2. Dataset

Many supermarket in Timor-Leste still lack of organizing or mining their day to day transaction data for further analysis due to lack mining data many supermarket still use traditional ways to create their marketing strategies, Association analysis and market basket analysis is very uncommon technology for these retailers to depend on to create their own marketing strategies, however this study is conducted in supermarket in Dili with dataset of 4 months transaction records from January to April involving total of 18818 rows of transaction and 4200 items , after generating all the association rules from the four period with apriori algorithm, the next analysis will overall variability of association rules for generate important or common rules that mostly occur in these four period. Typically, transactional datasets that include data on client purchases are used to perform market basket analysis. The columns reflect the things or goods that were purchased, and each row represents a transaction. An illustration of a transactional dataset is provided in Table 1-1.

The terms "items" and "transaction ID" in that collection refer to each transaction's unique identifier and each transaction's individual items, respectively. Any appropriate delimiter, such as commas, may be used to separate each item. A database or the CSV (comma-separated values) format are both acceptable ways to store the dataset. Make sure that the dataset has been cleaned up and preprocessed, eliminating any extra columns or rows and addressing missing or inconsistent information. Then, this dataset may be used to mine association rules and find item co-occurrence patterns using market basket analysis methods like Apriori or FP-Growth. It's crucial to keep in mind that this is a simplified example and that actual datasets for market basket research can contain hundreds or millions of transactions and a wide range of products.

3.4.3. Limitation

The main limitation of this work is that it doesn't have enough data to capture the seasonal pattern from the sales data. Here we have only used the 4-month data, and the 4 months only cover one season. So, there might be some products or items that have low sales and associations in these 4 months but have high sales and associations in the other half of the year. Besides this, this project focuses on the rules that are common in all four periods, which means it does not cover the seasonal associations between the products. For future work, it is preferred to have at least one year of data so that associations from all seasons can be explored and captured. and through that, it would be much easier to map out the most important and seasonal rules and how these rules vary with each other.

However, like any other dataset, retail stores also have certain limitations that can impact the analysis and insights derived from them. Some common limitations of retail store datasets include:

- **Sample Bias**

  Retail store datasets may not represent the entire customer population or may be skewed towards a specific demographic. This bias can limit the generalizability of the analysis and insights obtained from the dataset.

- **Missing Data**

  Datasets may contain missing or incomplete information, such as missing transaction records or incomplete customer profiles. Missing data can affect the accuracy and completeness of the analysis, potentially leading to biased results or limited insights[15].

- **Data Quality**

Data quality issues, such as errors, inconsistencies, or duplications, can arise in retail store datasets. These issues can introduce noise and inaccuracies into the analysis, impacting the reliability and validity of the findings.

- **Limited Contextual Information**

Retail store datasets may lack additional contextual information about customers, such as socio-economic factors, purchase motivations, or external events. The absence of such information can limit the depth of analysis and the ability to understand the underlying reasons or drivers behind customer behavior.

- **Data Volume and Timeframe**

The size and timeframe of the retail store dataset can impact the analysis. Small datasets may limit the scope and granularity of the analysis, while short timeframes may not capture long-term trends or seasonal effects.

- **Data Privacy and Anonymity**

Retail store datasets often contain sensitive customer information. Ensuring data privacy and anonymity while maintaining data utility can be a challenge, potentially limiting the availability or scope of certain data attributes for analysis.

- **External Factors:**

Retail store datasets may not account for external factors such as competitor actions, economic conditions, or marketing campaigns run by other businesses. Ignoring these external factors can limit the ability to understand and explain certain patterns or relationships in the data.

- **Data Integration**

  In some cases, retail store datasets may be fragmented or siloed, with data stored in different systems or formats. Integrating data from various sources can be challenging and may require additional preprocessing and data cleaning efforts.

It's important to consider these limitations when working with retail store datasets and interpreting the analysis results. Being aware of these constraints can help ensure a more accurate understanding of the insights derived from the data and enable informed decision-making.

3.4.4. Data Pre-Processing

In this section, will be describe the preprocessing of the dirty dataset from the supermarket until the data is ready for market basket analysis, when receiving the dataset from the supermarket the dataset is compost of 4 files of .xlsx format downloaded directly from the database of the supermarket from January to April, the next step, there are some data preprocessing techniques that need performs such as Data cleaning and data integration, bellow shows the required steps to producing the final dataset that ready for market basket analysis:

1. Selecting only relevant columns from each dataset for analysis
2. Exploring each Dataset
3. Merging all the dataset (January, February, March and April)
4. Improving all the columns name
5. Deleting all the missing values
6. Correcting Data types
7. Correcting the values of item description columns
8. Removing the columns that contain character that not applicable for analysis
9. Selecting only the data whose type is sale, and removing data whose types is return
10. Creating new features such as merging the month and date column

After successfully processing the dataset through all the steps above and the final dataset that is suitable for market basket analysis the next sections is data exploratory analysis.

3.4.5. Data Exploratory Analysis

Descriptive analysis and exploratory analysis are two crucial elements of data analysis that aid in comprehending and condensing datasets. Even though they have diverse functions, they are frequently combined to reveal new information about the data. Exploratory analysis: During the first stage of data analysis, the major objective is to examine and comprehend the dataset without having any preconceived ideas or preconceptions. It entails looking at the data from many perspectives, spotting trends, patterns, and linkages, and creating hypotheses for additional research. Visualizations, summary statistics, and data exploration approaches are frequently used in exploratory analysis. Here are a few typical methods for exploratory analysis: Summary figures: [16]Determine the mean, median, mode, standard deviation, and range of basic statistical metrics to comprehend the data's central tendency and variability.

On the other hand, a descriptive analysis entails summarizing and outlining a dataset's key characteristics. It concentrates on offering insightful analyses and summaries that accurately and succinctly convey the facts. The main goal is to convey the facts in an easily readable style. This analysis was implemented in a supermarket in Dili, Timor-Leste, involving a dataset of four months (January, February, March, and April) transaction records. The data consisted of 18818 total transactions with 4200 unique items sold. The association rule will be implemented for each period for each month; there were 4 periods for each month in this analysis. Table 2 shows the transaction data for each field.

Table 3-1 Number of transaction and item for each period

PERIOD	NUMBER OF TRANSACTION	NUMBER OF ITEM
1	1098	1350
2	6230	3556
3	6904	3730
4	4586	4200

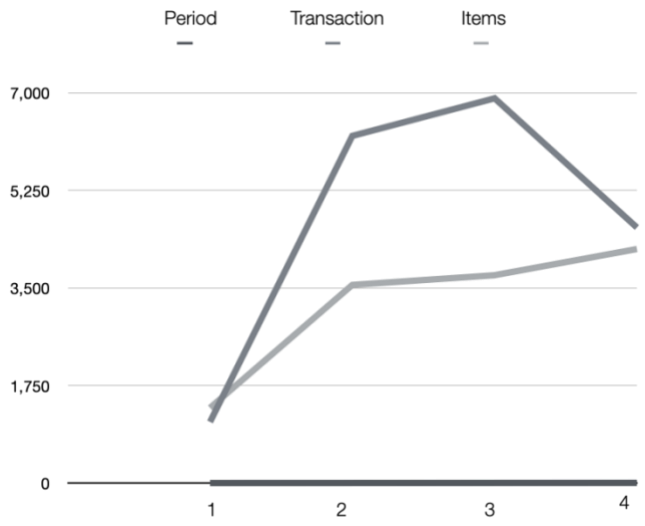

Figure 3-3 Chart of comparison between the number of transactions and items sold for each period

Table 1 shows the total number of unique items sold in each month and the total number of transactions that occurred in each month. Here we can see that the items sold in April belong to the maximum number of item types. This means that almost every transaction in April contains new and different items. Also in January, the number of unique items sold in January was 1350, and the total number of transactions was 1089. This means that, also in January, almost every single transaction contained a different product. However, in February and March, the total number of transactions is around 6,000, and the unique items sold are around 3000. This means that more similar products are sold in February and March. transactions occurred mostly in March, whereas

in January the least number of transactions occurred. The algorithm that was used to implement association rules was the Apriori algorithm, and Python is the programming language that is used to run the algorithm. This analysis used a dirty dataset that was downloaded with the Microsoft Excel extension, and before implementing the association rule to the dataset, the dataset had to undergo preprocessing that consisted of data cleaning, reduction, and integration. In this scenario, there is also a new column due to some technical problems in the different dates and times between the datasets of the four periods. After data pre-processing, the dataset for each period will be merged together for data exploratory analysis. The tables and figure below will show some statistics of the exploratory analysis, and the table below shows the clean dataset.

Table 3-2 Clean Dataset after data pre-processing and all combination of 4 dataset

No	Receipt_No	Date	Type	Item_description	Quantity	Revenue	month
1	202201280807449510001	2022-01-28 08:07:47	Sales	Spring Onion	0.46	1.15	1
2	202201280807449510001	2022-01-28 08:07:47	Sales	Spring Onion	0.126	0.32	1
3	202201280807449510001	2022-01-28 08:07:47	Sales	Spring Onion	0.162	0.41	1
4	202201280807449510001	2022-01-28 08:07:47	Sales	Spring Onion	0.162	0.41	1
	202201280807449510001	2022-01-28 08:07:47	Sales	Spring Onion	0.13	0.33	1
11	202201280810176100003	2022-01-28 08:10:18	Sales	Aqua 600ml	1.0	0.25	1
14	202201280811223250004	2022-01-28 08:11:24	Sales	Mie Sedaap White Curry 81g	1.0	0.25	1
17	202201280812336110005	2022-01-28 08:12:36	Sales	ToraBika Cappuccino 250g	1.0	1.5	1
20	202201280814337510006	2022-01-28 08:14:34	Sales	Aqua 1500ml	1.0	0.5	1
23	202201280815204540007	2022-01-28 08:15:22	Sales	Aqua 600ml	1.0	0.25	1
…	……	……	……	……	…	…	…
17863	202204301858066170193	2022-04-30 18:58:08	Sales	Non-Woven Bag PcsM	1.0	0.15	4

Table 3-3 20 Most Frequently sold items for all period

No	Items	frequency
1	Aqua 1500ml(1x12x1500ml)	2123
2	Non-Woven Bag PcsS	1635
3	Non-Woven Bag PcsM	1465
4	Aqua 600ml(1x24x600ml)	1460
5	Chicken Leg Quarter	1255
6	Ikan Kembong	1007
7	Non-Woven Bag PcsL	506
8	Le Minerale 600ml	476
9	Weizhang Cake 1.5	359
10	Bawang Putih	353
11	Sosis Perdix 340g	338
12	Bokchoy	334
13	Le Minerale 1500ml	310
14	Sedaap Goreng 90g	297
15	Bawang Merah Kecil	278
16	Apple Fuji	265
17	Aqua 330ml	264
18	Pucuk Harum 350ml	260
19	Apple Pink Lady	245
20	Aqua Gota 1500ml	244

The above plot shows the top 20 items that are rarely sold during the 4-month period. As the products that are only sold once or twice over the 4-month period, we should remove them from our analysis and try not to add any value to our analysis.

Table 5-4 20 least sold items for all period

No	**Items**	**frequency**
1	Biore BF 600ml Lively Refresh	1
2	Campbell's Soup Chunky Chicken Vegetable 505g(1x12x505g)	1
3	Campbell's Soup CL Creamy Chicken 500g(1x12x500g)	1
4	Nobre Naturissimos Salcichas 160g	1
5	Mega Sardines In Natural Oil 155g(1x48x155g)	1
6	Edgell 4x125g Super Sweet Corn	1
7	Emeron Hair vitamin Soft & Smoot	1
8	SmartHeart Dog Creamy Treat Adult Chicken and Pumpkin 60gr	1
9	Ovale FL Yam Beam 100ml	1
10	Original Source Shower Gel Vanilla & Raspberry 250ml	1
11	211120R-6 Suprfine Fiber Twist Mop 03(1x50)	1
12	Briko Chocolate 125g	1
13	Koepoe TBM 80g	1
14	Pizza Pepperoni 350g	1
15	17708D-9 Kemei Barber Scissors KM-025(1x60)	1
16	SumberAyu Daun Sirih 90ml Musk	1
17	Tiga Roda 8Jam	1
18	Palmolive 180ml Silky Straight	1
19	SmartHeart Dog Creamy Treat Adult Chicken and Spinach 60g	1
20	21L33 Weilong Spicy Stick 50g	1

To find the most important items that should always be in store, we will first filter out the top 30 most commonly sold products and check if they sell every month; if they do, that means they are the items that should always be present in the store. That is how we can detect the most important items in terms of selling patterns.

For this, we first need to create a week column to show us the items that always appear twice or three times in a month, like the table below:

Table 3-5. Items that always appear twice to four times in a month

No	Item_description	month	total_month
1	Apple Fuji	[1 2 3 4]	4
2	Apple Pink Lady	[1 2 3 4]	4
3	Aqua 1500ml(1x12x1500ml)	[2 3 4]	3
4	Aqua 330ml	[1 2 3 4]	4
5	Aqua 600ml(1x24x600ml)	[2 3 4]	3
6	Aqua Fresca 1500ml(1x12x1500ml)	[2 3 4]	3
7	Aqua Gota 1500ml	[1 2 3 4]	4
8	Aqua Gota 600ml	[1 2 3 4]	4
9	Bawang Merah Kecil	[1 2 3 4]	4
10	Bawang Putih	[1 2 3 4]	4
11	Bokchoy	[1 2 3 4]	4
12	Chicken Leg Quarter/ Paha Besar	[1 2 3 4]	4
13	Chinese Cabbage	[1 2 3 4]	4
14	Country Choice 250ml Jambu(1x24)	[2 3 4]	3
15	Fresh Orange Valencia 72s/88s	[1 2 3 4]	4
16	Ikan Kembong	[1 2 3 4]	4
17	Kentang/Potato Timbang	[1 2 3 4]	4
18	Le Minerale 1500ml	[1 2 3 4]	4
19	Le Minerale 600ml	[1 2 3 4]	4
20	Non-Woven Bag PcsL	[1 2 3 4]	4
21	Non-Woven Bag PcsM	[1 2 3 4]	4
22	Non-Woven Bag PcsS	[1 2 3 4]	4
23	Pork Belly/Perut	[1 2 3 4]	4
24	Pucuk Harum 350ml	[1 2 3 4]	4
25	R.Marlboro Putih	[1 2 3 4]	4
26	Sedaap Goreng 90g	[1 2 3 4]	4
27	Sosis Borella 340g	[1 2 3 4]	4
28	Sosis Perdix 340g	[1 2 3 4]	4
29	Weizhang Cake 1.5	[1 2 3 4]	4
30	You C1000 Orange 140ml	[2 3 4]	3

After having the list of items that mostly appear every week, we can now shorten the list to find the item that is always present in the sale every week. The table below shows the items that are sold every week.

Table 3-6 25 items that always appear every week.

No	Item_description	month	total_month
1	Apple Fuji	[1 2 3 4]	4
2	Apple Pink Lady	[1 2 3 4]	4
3	Aqua 330ml	[1 2 3 4]	4
4	Aqua Gota 1500ml	[1 2 3 4]	4
5	Aqua Gota 600ml	[1 2 3 4]	4
6	Bawang Merah Kecil	[1 2 3 4]	4
7	Bawang Putih	[1 2 3 4]	4
8	Bokchoy	[1 2 3 4]	4
9	Chicken Leg Quarter/ Paha Besar	[1 2 3 4]	4
10	Chinese Cabbage	[1 2 3 4]	4
11	Fresh Orange Valencia 72s/88s	[1 2 3 4]	4
12	Ikan Kembong	[1 2 3 4]	4
13	Kentang/Potato Timbang	[1 2 3 4]	4
14	Le Minerale 1500ml	[1 2 3 4]	4
15	Le Minerale 600ml	[1 2 3 4]	4
16	Non-Woven Bag PcsL	[1 2 3 4]	4
17	Non-Woven Bag PcsM	[1 2 3 4]	4
18	Non-Woven Bag PcsS	[1 2 3 4]	4
19	Pork Belly/Perut	[1 2 3 4]	4
20	Pucuk Harum 350ml	[1 2 3 4]	4
21	R.Marlboro Putih	[1 2 3 4]	4
22	Sedaap Goreng 90g	[1 2 3 4]	4
23	Sosis Borella 340g	[1 2 3 4]	4
24	Sosis Perdix 340g	[1 2 3 4]	4
25	Weizhang Cake 1.5	[1 2 3 4]	4

After we have the list of items that always appear in the store, we can see the top 5 items that are always present in the store during four different periods.

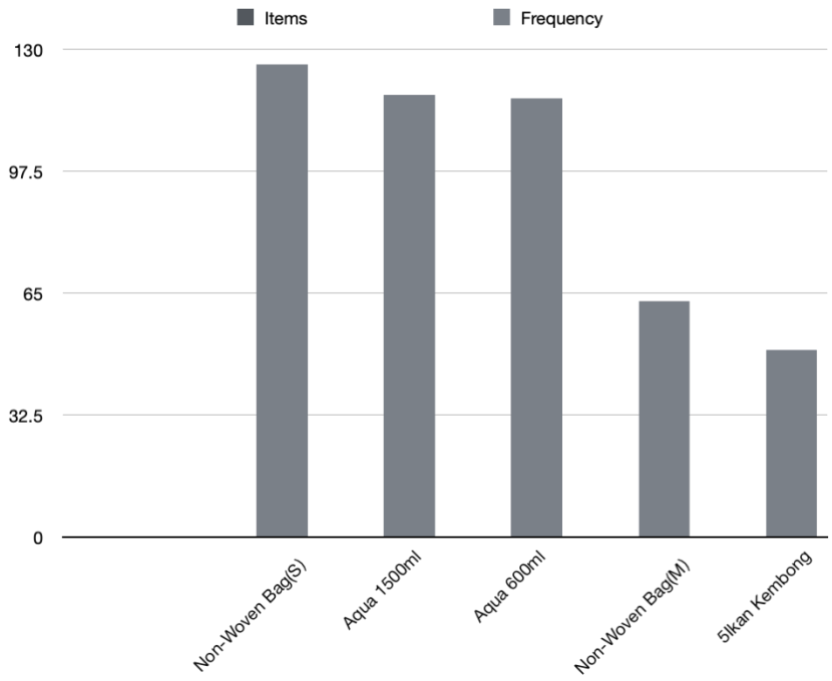

Figure 3-4 Top 5 items sold in period 1

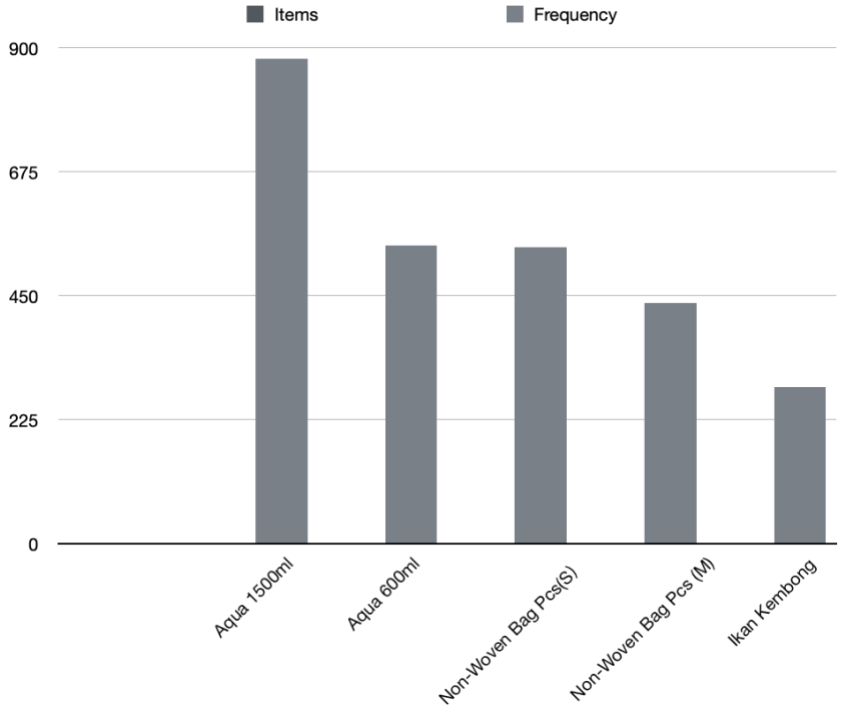

Figure 3-5 Top 5 items sold in period 2

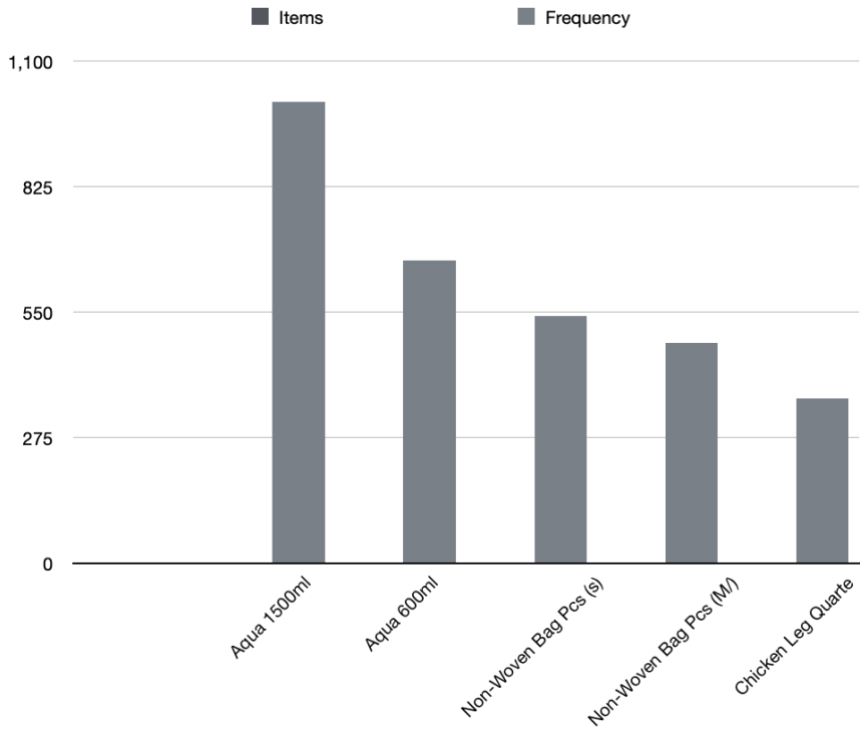

Figure 3-6 Top 5 items sold in period 3

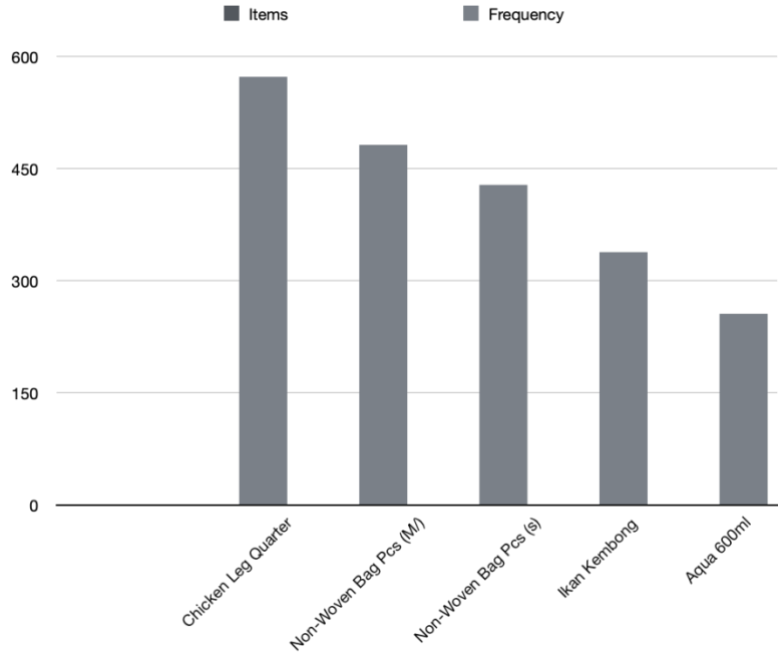

Figure 3-7 Top 5 items sold in period 4

Figure 3-8. The above plot shows the top 30 products or items that are mostly sold

The purpose of the exploratory analysis is to make some statistical summaries and get a better understanding of the dataset.

3.4.6. Association Mining Rule for Market Basket Analysis

After analyzing each product individually, the next step will be the process to analyze the relationship between these products, commonly known as market basket analysis (MBA). First, we create a basket of products per id_number, then we use the Transaction Encoder object to transform this dataset into an array format suitable for typical machine APIs. Via the fit method, the transaction encoder learns the unique labels in the dataset, and via the transform method, it transforms the input dataset (a Python list of lists) into a one-hot encoded NumPy Boolean array. Then I converted it into a data frame. Then generate the frequent item set (products that are frequently bought together) with a chosen support threshold value. This support threshold value is used to choose the product subsets that have most frequently occurred in the dataset. We set the minimum support threshold value to 0.003, which means the occurrence of the product subset in the observed dataset should be at least 3%. After determining the threshold value on the basis of these frequent items, we define rules that indicate the chances of the occurrence of the product subset in the entire transaction. In other words, we can say that rules define how the product subset

will occur in the future. Rules does this on the basis of support, confidence, and lift values. The higher the support, confidence, and lift value, the higher the chance of the product subset re-occurrence[17].

3.4.7. Create Item Matrix

A key component of association rule analysis for market basket analysis is the creation of an item matrix. The existence or absence of items in each transaction is captured by the item matrix, which acts as a data representation, allowing associations to be found and useful rules to be created. The following are the primary purposes of developing an item matrix in association rules analysis for market basket analysis:

- **Transaction Encoding**

The item matrix represents a tabular structure where each row corresponds to a transaction and each column corresponds to an item. By encoding transactions into the item matrix, the presence or absence of items in each transaction is captured, forming a binary representation. This encoding is essential for subsequent analysis and rule generation

- **Frequency Calculation**

The item matrix allows for the calculation of item frequencies, which indicate how often each item appears across all transactions. These frequencies are essential for identifying frequent item sets, which are sets of items that occur together above a predefined minimum support threshold. The item matrix facilitates the counting and aggregation of item occurrences, enabling the determination of item frequencies.

- **Frequency Calculation**

The item matrix allows for the calculation of item frequencies, which indicate how often each item appears across all transactions. These frequencies are essential for identifying frequent item sets, which are sets of items that occur together above a predefined minimum support threshold. The item matrix facilitates the counting and aggregation of item occurrences, enabling the determination of item frequencies.

3.4.8. Generate Association Rules

The item matrix serves as the basis for discovering association rules. It enables the identification of item sets that meet the minimum support threshold and subsequently generates association rules based on these item sets. The item matrix provides the necessary information to evaluate item co-occurrences and calculate confidence measures for rule generation frequencies. The points below will describe some important metrics for association rules generation.

1. support is used to measure the abundance or frequency (often interpreted as significance or importance) of an item set in a dataset. its value range is 0 to 1. here because of huge numbers of product we can see that support values are below 0.1.
2. The confidence of a rule is the probability of seeing the consequent in a transaction given that it also contains the antecedent. it ranges from 0 to 1.
3. The lift metric is commonly used to measure how much more often the antecedent and consequent of a rule occur together than we would expect if they were statistically independent. If antecedent and consequent are independent, the Lift score will be exactly 1.
4. Leverage computes the difference between the observed frequency of A and C appearing together and the frequency that would be expected if A and C were independent. A leverage value of 0 indicates independence.
5. A high conviction value means that the consequent is highly depending on the antecedent.

3.4.9. Define Important Rules With OCVR

After get the association rules, the next step is to used overall variability of association rules to find common rules most frequently appear rules in each month as most important rules to create marketing strategy.

Chapter 4: Results

4.1. Association Rules Generated for Each Period

The most frequent item for each period was examined as initial information before implanting AR in the transaction data. The most frequent item for each period is similar, as indicated in the charts. It includes a number of things, like a water bottle of 1500 milliliters, a chicken leg, fish, etc.

Table 4-1 Frequent items that appear in all period

Period 1		**Period 2**	
Items	**frequency**	**Items**	**frequency**
Non-Woven Bag/pcs（S）	126	Non-Woven Bag/Pcs（S）	539
Aqua 1500ml	118	Aqua 1500ml(1x12x1500ml)	880
Aqua 600ml	117	Aqua 600ml(1x24x600ml)	541
Non-Woven Bag/Pcs（M）	63	Non-Woven Bag/Pcs（M）	437
Ikan Kembong 鲭鱼	50	Ikan Kembong 鲭鱼	284
Chicken Leg Quarter/ 鸡大腿	45	Chicken Leg Quarter/鸡大腿	275

Period 3		**Period 4**	
Items	**frequency**	**Items**	**frequency**
Non-Woven Bag/Pcs（S）	542	Non-Woven Bag/Pcs（S）	428
Aqua 1500ml(1x12x1500ml)	1012	Aqua 1500ml(1x12x1500ml)	231
Aqua 600ml(1x24x600ml)	663	Aqua 600ml(1x24x600ml)	256
Non-Woven Bag/Pcs（M）	483	Non-Woven Bag/Pcs（M）	482
Ikan Kembong 鲭鱼	335	Ikan Kembong 鲭鱼	338
Chicken Leg Quarter/鸡大腿	362	Chicken Leg Quarter/鸡大腿	573

The minimum threshold value for support is 0.01 and the confidence is 0.3, which will be used to implement association rules using the apriori algorithm to generate rules. Tables 6, 7, 8, and 9 will

show the rules generated for each period, and the tables below will show the rules generated for period 1 until period 4.

Table 4-2 Rules generated for Period 1

No	antecedents	consequents	support	confidence	lift
1	Apple Fuji	Bokchoy	0.00331858	0.176471	7.251337
2	Bokchoy	Apple Fuji	0.00331858	0.136364	7.251337
3	Ikan Kembong	Apple Fuji	0.00331858	0.15	7.976471
4	Apple Fuji	Ikan Kembong	0.00331858	0.176471	7.976471
5	Apple Fuji	Non-Woven Bag PcsM	0.00331858	0.176471	2.532213
6	Fresh Grape Red China (22-26cm)	Apple Pink Lady	0.00663717	0.315789	16.79257
7	Apple Pink Lady	Fresh Grape Red China (22-26cm)	0.00663717	0.352941	16.79257
8	Non-Woven Bag PcsL	Apple Pink Lady	0.00553097	0.151515	8.057041
9	Apple Pink Lady	Non-Woven Bag PcsL	0.00553097	0.294118	8.057041
10	Apple Pink Lady	Non-Woven Bag PcsM	0.00442478	0.235294	3.376284
11	Apple Pink Lady	Non-Woven Bag PcsS	0.00331858	0.176471	1.266106
12	Le Minerale 1500ml	Aqua 1500ml	0.00553097	0.172414	1.320865
13	Nice Facial Tissue 2ply 60s	Aqua 1500ml	0.00331858	0.428571	3.283293
14	Pucuk Harum 350ml	Aqua 600ml	0.00331858	0.428571	3.311355
15	Aurora Chicken Half Breast 2kg	Telur no2 PACK	0.00331858	0.6	67.8
16	Telur no2 PACK	Aurora Chicken Half Breast 2kg	0.00331858	0.375	67.8
17	Ayam 1200g Sadia	Ayam 1400g Sadia	0.00331858	0.6	180.8
18	Ayam 1400g Sadia	Ayam 1200g Sadia	0.00331858	1	180.8
19	Ayam 1200g Sadia	Bokchoy	0.00331858	0.6	24.65455
20	Bokchoy	Ayam 1200g Sadia	0.00331858	0.136364	24.65455
21	Bawang Putih	Bawang Merah Kecil	0.00995575	0.428571	32.28571
22	Bawang Merah Kecil	Bawang Putih	0.00995575	0.75	32.28571
23	Bokchoy	Bawang Merah Kecil	0.00331858	0.136364	10.27273
24	Bawang Merah Kecil	Bokchoy	0.00331858	0.25	10.27273
25	Kentang/Potato Timbang	Bawang Merah Kecil	0.00331858	0.428571	32.28571
26	Bawang Merah Kecil	Kentang/Potato Timbang	0.00331858	0.25	32.28571
27	Bawang Merah Kecil	Non-Woven Bag PcsM	0.00442478	0.333333	4.783069
28	Telur no4 PACK	Bawang Merah Kecil	0.00331858	0.5	37.66667
29	Bawang Merah Kecil	Telur no4 PACK	0.00331858	0.25	37.66667
30	Non-Woven Bag PcsM	Bawang Putih	0.00995575	0.142857	6.14966

31	Bawang Putih	Non-Woven Bag PcsM	0.00995575	0.428571	6.14966
32	Carrot/Wortel	Bokchoy	0.00442478	0.5	20.54545
33	Bokchoy	Carrot/Wortel	0.00442478	0.181818	20.54545
34	Fresh Orange Valencia 72s/88s	Bokchoy	0.00553097	0.227273	9.338843
35	Bokchoy	Fresh Orange Valencia 72s/88s	0.00553097	0.227273	9.338843
36	Non-Woven Bag PcsL	Bokchoy	0.00553097	0.151515	6.225895
37	Bokchoy	Non-Woven Bag PcsL	0.00553097	0.227273	6.225895
38	Non-Woven Bag PcsM	Bokchoy	0.00774336	0.111111	4.565657
39	Bokchoy	Non-Woven Bag PcsM	0.00774336	0.318182	4.565657
40	Bokchoy	Non-Woven Bag PcsS	0.00442478	0.181818	1.304473
41	Telur no2 PACK	Bokchoy	0.00442478	0.5	20.54545
42	Bokchoy	Telur no2 PACK	0.00442478	0.181818	20.54545
43	Telur no4 PACK	Bokchoy	0.00331858	0.5	20.54545
44	Bokchoy	Telur no4 PACK	0.00331858	0.136364	20.54545
45	Buavita 250ml Guava	Non-Woven Bag PcsM	0.00442478	0.166667	2.391534
46	Buavita 250ml Guava	Non-Woven Bag PcsS	0.00442478	0.166667	1.195767
47	Cadbury 65g Cashew Nut	Non-Woven Bag PcsS	0.00331858	0.3	2.152381
48	Chicken Leg Quarter/ Paha Besar	Non-Woven Bag PcsM	0.00663717	0.352941	5.064426
49	Chicken Leg Quarter/ Paha Besar	Non-Woven Bag PcsS	0.00331858	0.176471	1.266106
50	Fresh Grape Red China (22-26cm)	Fresh Orange Valencia 72s/88s	0.00442478	0.210526	8.650718
51	Fresh Orange Valencia 72s/88s	Fresh Grape Red China (22-26cm)	0.00442478	0.181818	8.650718
52	Fresh Grape Red China (22-26cm)	Fresh Pear Ya / ya li	0.00331858	0.157895	28.54737
53	Fresh Pear Ya /ya li	Fresh Grape Red China (22-26cm)	0.00331858	0.6	28.54737
54	Fresh Grape Red China (22-26cm)	Non-Woven Bag PcsL	0.00442478	0.210526	5.767145
55	Non-Woven Bag PcsL	Fresh Grape Red China (22-26cm)	0.00442478	0.121212	5.767145
56	Fresh Grape Red China (22-26cm)	Non-Woven Bag PcsM	0.00774336	0.368421	5.28655
57	Non-Woven Bag PcsM	Fresh Grape Red China (22-26cm)	0.00774336	0.111111	5.28655
58	Fresh Grape Red China (22-26cm)	Non-Woven Bag PcsS	0.00442478	0.210526	1.510443
59	Fresh Orange Valencia 72s/88s	Non-Woven Bag PcsL	0.00331858	0.136364	3.735537

60	Fresh Orange Valencia 72s/88s	Non-Woven Bag PcsM	0.00442478	0.181818	2.608947
61	Fresh Orange Valencia 72s/88s	Telur no2 PACK	0.00442478	0.181818	20.54545
62	Telur no2 PACK	Fresh Orange Valencia 72s/88s	0.00442478	0.5	20.54545
63	Ikan Kembong	Non-Woven Bag PcsM	0.00663717	0.3	4.304762
64	Ikan Kembong	Non-Woven Bag PcsS	0.00331858	0.15	1.07619
65	Non-Woven Bag PcsL	Jagua Chicken Middle Joint Wing 1kg	0.00442478	0.121212	27.39394
66	Jagua Chicken Middle Joint Wing 1kg	Non-Woven Bag PcsL	0.00442478	1	27.39394
67	Non-Woven Bag PcsL	Non-Woven Bag PcsM	0.00884956	0.242424	3.478595
68	Non-Woven Bag PcsM	Non-Woven Bag PcsL	0.00884956	0.126984	3.478595
69	Non-Woven Bag PcsL	Pork Belly/Perut	0.00442478	0.121212	9.131313
70	Pork Belly/Perut	Non-Woven Bag PcsL	0.00442478	0.333333	9.131313
71	Short Necked / Flower Clam 500g	Non-Woven Bag PcsL	0.00331858	0.6	16.43636
72	Telur no4 PACK	Non-Woven Bag PcsL	0.00331858	0.5	13.69697
73	Yakult 65ml 5s	Non-Woven Bag PcsL	0.00331858	0.166667	4.565657
74	Pork Belly/Perut	Non-Woven Bag PcsM	0.00331858	0.25	3.587302
75	Sosis Borella 340g	Non-Woven Bag PcsM	0.00442478	0.4	5.739683
76	Sosis Perdix 340g	Non-Woven Bag PcsM	0.00663717	0.5	7.174603
77	Telur no4 PACK	Non-Woven Bag PcsM	0.00331858	0.5	7.174603
78	Weizhang Cake 1.5	Non-Woven Bag PcsM	0.00774336	0.388889	5.580247
79	Non-Woven Bag PcsM	Weizhang Cake 1.5	0.00774336	0.111111	5.580247
80	Yakult 65ml 5s	Non-Woven Bag PcsM	0.00553097	0.277778	3.985891
81	Sosis Borella 340g	Non-Woven Bag PcsS	0.00331858	0.3	2.152381
82	Sweet Cake $1.5	Non-Woven Bag PcsS	0.00331858	0.375	2.690476
83	Weizhang Cake 1.5	Non-Woven Bag PcsS	0.00331858	0.166667	1.195767
84	You C1000 Lemon 140ml	Non-Woven Bag PcsS	0.00331858	0.3	2.152381
85	You C1000 Lemon 140ml	You C1000 140ml Orange	0.00331858	0.3	18.08
86	You C1000 140ml Orange	You C1000 Lemon 140ml	0.00331858	0.2	18.08

Table 4-3 Rules generated for Period 2

No	antecedents	consequents	support	confidence	lift
1	Country Choice 250ml J	Aqua 1500ml(1x12x1500ml)	0.00309	0.17	1.062886
2	Le Minerale 1500ml	Aqua 1500ml(1x12x1500ml)	0.00418	0.133721	0.83606
3	You C1000140ml Orange	Aqua 1500ml(1x12x1500ml)	0.00418	0.205357	1.283949
4	Ayam 1200g Sadia	Ayam 1400g Sadia	0.003817	0.875	192.57
5	Ayam 1400g Sadia	Ayam 1200g Sadia	0.003817	0.84	192.57

6	Ayam 1200g Sadia	Dodo Fish Ball 200g	0.00309	0.708333	149.8942
7	Dodo Fish Ball 200g	Ayam 1200g Sadia	0.00309	0.653846	149.8942
8	Fresh Orange Valencia	Ayam 1200g Sadia	0.003453	0.253333	58.07667
9	Ayam 1200g Sadia	Fresh Orange Valencia	0.003453	0.791667	58.07667
10	Telur no2 PACK	Ayam 1200g Sadia	0.003453	0.387755	88.89286
11	Ayam 1200g Sadia	Telur no2 PACK	0.003453	0.791667	88.89286
12	Dodo Fish Ball 200g	Ayam 1400g Sadia	0.003272	0.692308	152.3631
13	Ayam 1400g Sadia	Dodo Fish Ball 200g	0.003272	0.72	152.3631
14	Fresh Orange Valencia	Ayam 1400g Sadia	0.003817	0.28	61.6224
15	Ayam 1400g Sadia	Fresh Orange Valencia	0.003817	0.84	61.6224
16	Ayam 1400g Sadia	Rimbaria Saos 600ml	0.00309	0.68	196.9137
17	Rimbaria Saos 600ml	Ayam 1400g Sadia	0.00309	0.894737	196.9137
18	Telur no2 PACK	Ayam 1400g Sadia	0.003635	0.408163	89.82857
19	Ayam 1400g Sadia	Telur no2 PACK	0.003635	0.8	89.82857
20	Bawang Merah Kecil	Bawang Putih	0.005816	0.4	24.18462
21	Bawang Putih	Bawang Merah Kecil	0.005816	0.351648	24.18462
22	Bawang Putih	Non-Woven Bag PcsM	0.00418	0.252747	3.182186
23	Fresh Orange Valencia	Bokchoy	0.003272	0.24	15.17793
24	Bokchoy	Fresh Orange Valencia	0.003272	0.206897	15.17793
25	Chicken Leg Quarter	Non-Woven Bag PcsM	0.005816	0.304762	3.837071
26	Chicken Leg Quarter	Non-Woven Bag PcsS	0.00418	0.219048	2.235993
27	Chicken Leg Quarter	Sosis Perdix 340g	0.00418	0.219048	12.82128
28	Sosis Perdix 340g	Chicken Leg Quarter	0.00418	0.244681	12.82128
29	Fresh Orange Valencia	Dodo Fish Ball 200g	0.003272	0.24	50.78769
30	Dodo Fish Ball 200g	Fresh Orange Valencia	0.003272	0.692308	50.78769
31	Telur no2 PACK	Dodo Fish Ball 200g	0.00309	0.346939	73.41758
32	Dodo Fish Ball 200g	Telur no2 PACK	0.00309	0.653846	73.41758
33	Fresh Orange Valencia	Mie Atoom Bulan 200g	0.00309	0.226667	56.68727
34	Mie Atoom Bulan 200g	Fresh Orange Valencia	0.00309	0.772727	56.68727
35	Telur no2 PACK	Fresh Orange Valencia	0.003453	0.387755	28.44571
36	Fresh Orange Valencia	Telur no2 PACK	0.003453	0.253333	28.44571
37	Ikan Kembong	Non-Woven Bag PcsM	0.005453	0.283019	3.563318
38	Ikan Kembong	Non-Woven Bag PcsS	0.003817	0.198113	2.022298
39	Sosis Borella 340g	Non-Woven Bag PcsM	0.003999	0.44898	5.652828
40	Sosis Perdix 340g	Non-Woven Bag PcsM	0.004726	0.276596	3.482448
41	Weizhang Cake 1.5	Non-Woven Bag PcsM	0.00418	0.255556	3.217544

Table 4-4 Rules generated for Period 3

NO	antecedents	consequents	support	confidence	lift
1	Apple Fuji	Non-Woven Bag PcsM	0.003079	0.162393	2.074463
2	Apple Pink Lady	Non-Woven Bag PcsM	0.00389	0.333333	4.258109
3	Aqua 600ml(1x24x600ml)	Aqua 1500ml(1x12x1500ml)	0.0094	0.087481	0.533358
4	Ultramilk Coklat 125ml	Aqua 1500ml(1x12x1500ml)	0.004052	0.304878	1.858792
5	Ayam 1200g Sadia	Dodo Fish Ball 200g(1x30x200g)	0.003241	0.740741	190.4321
6	Dodo Fish Ball 200g(1x30x200g)	Ayam 1200g Sadia	0.003241	0.833333	190.4321
7	Bawang Putih	Bawang Merah Kecil	0.005673	0.321101	30.95614
8	Bawang Merah Kecil	Bawang Putih	0.005673	0.546875	30.95614
9	Bawang Putih	Non-Woven Bag PcsM	0.00389	0.220183	2.812696
10	Chicken Leg Quarter/ Paha Besar	Ikan Kembong	0.005511	0.300885	11.53081
11	Ikan Kembong	Chicken Leg Quarter/ Paha Besar	0.005511	0.21118	11.53081
12	Chicken Leg Quarter/ Paha Besar	Non-Woven Bag PcsM	0.00389	0.212389	2.713131
13	Chicken Leg Quarter/ Paha Besar	Non-Woven Bag PcsS	0.003241	0.176991	2.014825
14	Chicken Leg Quarter/ Paha Besar	Sosis Borella 340g	0.003079	0.168142	12.2051
15	Sosis Borella 340g	Chicken Leg Quarter/ Paha Besar	0.003079	0.223529	12.2051
16	Chicken Leg Quarter/ Paha Besar	Sosis Perdix 340g	0.0047	0.256637	14.39501
17	Sosis Perdix 340g	Chicken Leg Quarter/ Paha Besar	0.0047	0.263636	14.39501
18	Ikan Kembong	Non-Woven Bag PcsM	0.006483	0.248447	3.173746
19	Non-Woven Bag PcsM	Ikan Kembong	0.006483	0.082816	3.173746
20	Ikan Kembong	Non-Woven Bag PcsS	0.004052	0.15528	1.767665
21	Sosis Borella 340g	Ikan Kembong	0.0047	0.341176	13.0749
22	Ikan Kembong	Sosis Borella 340g	0.0047	0.180124	13.0749
23	Sosis Perdix 340g	Ikan Kembong	0.003241	0.181818	6.967815
24	Ikan Kembong	Sosis Perdix 340g	0.003241	0.124224	6.967815
25	Sosis Borella 340g	Non-Woven Bag PcsM	0.00389	0.282353	3.606869
26	Sosis Perdix 340g	Non-Woven Bag PcsM	0.004538	0.254545	3.251647

Table 4-5 Rules generated for Period 4

No	antecedents	consequents	support	Confidence	lift
1	Bawang Merah Kecil	Bawang Putih	0.017866	0.575	18.66092
2	Bawang Putih	Bawang Merah Kecil	0.017866	0.579832	18.66092
3	Bawang Merah Kecil	Non-Woven Bag PcsM	0.007768	0.25	2.003112
4	Bawang Putih	Chicken Leg Quarter	0.005697	0.184874	4.434678
5	Chicken Leg Quarter	Bawang Putih	0.005697	0.136646	4.434678
6	Bawang Putih	Kentang/Potato Timbang	0.006214	0.201681	7.788908
7	Kentang/Potato Timbang	Bawang Putih	0.006214	0.24	7.788908
8	Bawang Putih	Non-Woven Bag PcsL	0.005697	0.184874	2.987377
9	Bawang Putih	Non-Woven Bag PcsM	0.007509	0.243697	1.952613
10	Bokchoy	Non-Woven Bag PcsL	0.005438	0.184211	2.976657
11	Bokchoy	Non-Woven Bag PcsM	0.005697	0.192982	1.546262
12	Ikan Kembong	Chicken Leg Quarter	0.009063	0.257353	6.173274
13	Chicken Leg Quarter	Ikan Kembong	0.009063	0.217391	6.173274
14	Chicken Leg Quarter	Non-Woven Bag PcsM	0.008545	0.204969	1.642303
15	Chicken Leg Quarter	Non-Woven Bag PcsS	0.005179	0.124224	1.120915
16	Chicken Leg Quarter	Sosis Borella 340g	0.005697	0.136646	6.136357
17	Sosis Borella 340g	Chicken Leg Quarter	0.005697	0.255814	6.136357
18	Chicken Leg Quarter	Sosis Perdix 340g	0.008545	0.204969	6.488443
19	Sosis Perdix 340g	Chicken Leg Quarter	0.008545	0.270492	6.488443
20	Ikan Kembong	Non-Woven Bag PcsM	0.008027	0.227941	1.826367
21	Ikan Kembong	Sosis Perdix 340g	0.005179	0.147059	4.655256
22	Sosis Perdix 340g	Ikan Kembong	0.005179	0.163934	4.655256
23	Non-Woven Bag PcsL	Pork Belly/Perut	0.006214	0.100418	5.540227
24	'Pork Belly/Perut	Non-Woven Bag Pcs	0.006214	0.342857	5.540227
25	{'Non-Woven Bag PcsL	Sosis Perdix 340g	0.006214	0.100418	3.178819
26	Sosis Perdix 340g	Non-Woven Bag PcsL	0.006214	0.196721	3.178819
27	Non-Woven Bag PcsL	Weizhang Cake 1.	0.00725	0.117155	2.513622
28	Weizhang Cake 1.5	Non-Woven Bag PcsL	0.00725	0.155556	2.513622
29	Sosis Borella 340g	Non-Woven Bag PcsM	0.005955	0.267442	2.142864
30	Sosis Perdix 340g	Non-Woven Bad PcsM	0.00725	0.229508	1.838923
31	Weizhang Cake 1.5	'Non-Woven Bag PcsM	0.009839	0.211111	1.691517

The table below show the total rule get from each period and a brief comparison of the support and confidence for each period. In the first period, the second period, and the third period, the minimum support value is set to 0.003, and the minimum confidence value is 0.1 for the first period

and second period and 0.08 for the third period. However, for the last period, the minimum support value is 0.005 and the minimum confidence value is 0.10. With these metrics, the first period successfully generated 86 rules, the second period 41 rules, the third period 26 rules, and the fourth period 31 rules.

Table 4-6 Comparison of support and confidence minimum thresholds for each period

period	**Min_Support**	**Min_Confidence**	**Number of Rules**
1	0.003	0.1	86
2	0.003	0.1	41
3	0.003	0.08	26
4	0.005	0.1	31
Total Rules			184

The rules generated from each period were quite different; further research is needed to create a better marketing strategy. OCVR, or overall variability of association rule analysis, is the next step to knowing any important rules in each period.

To compare the support and confidence thresholds for different periods, you would need to analyze the dataset for each period separately and evaluate the impact of different threshold values on the generated association rules. This analysis would require specific data and context to provide meaningful comparisons. The determination of support and confidence thresholds in market basket analyses depends on the particular objectives and specifications of the study as well as the features of the dataset under consideration. The confidence threshold establishes the minimal degree of certainty required for an association rule to be deemed significant, whereas the support threshold establishes the minimal frequency or occurrence of an item in the dataset[7].

A higher support criterion typically results in a smaller number of potentially relevant correlations because only item sets with a greater frequency of occurrence will be taken into account. On the other hand, a lower support level will enable more item sets, including less frequent ones, to be taken into account.

The confidence threshold establishes the bare minimum of accuracy or dependability necessary for an association rule to be regarded as relevant. A lower threshold will provide a broader range of rules with varied degrees of confidence, whereas a higher threshold will produce a smaller number of association rules that are very confident. A trade-off between the thoroughness and specificity of the association rules often occurs when choosing support and confidence criteria. A higher threshold ensures more reliable and significant rules but may miss some potentially interesting associations, while a lower threshold may result in a larger number of less significant or noisy rules.

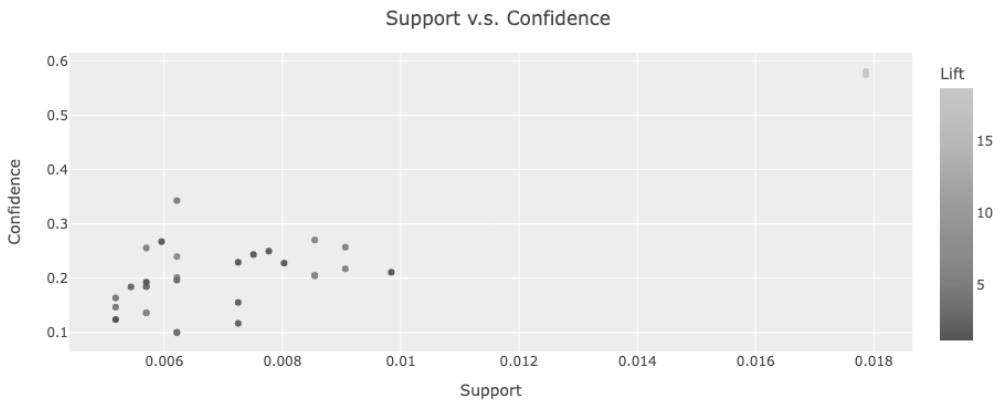

Figure 4-1 shows the comparison between the support and confidence of all combinations.

1.2. Rules generated With Overall Variability of Association Rule (OCVR)

To get the Overall Variety of Association rules values, required to generate an association rule in each period. The first step is to combine rules that mostly appear in each period, while rules that did not appear in each period were not used, there were 8 rules that mostly appear in each period. To find OCVR, first step is to calculate index variability lift (CVL) and index variability confidence (CVC) with the value of confidence and lift ratio in each period. In table 11 shows the calculation result of overall variability of association rule. These values show low variability according to [7] (2011) are 1% to 30%. OCVR value greater than 30% indicates that the rule is very

vulnerable to change and cannot be used at any time. The result shows that OCVR value or all rules were 1% < OCVR < 30%, it means all rules can be used to make marketing strategies.

Table 4-7 The overall variability of association rules

No	antecedents	consequents	CVC	CVL	OCVR
1	Bawang Merah Kecil	Bawang Putih	0.252791	0.238607	0.245699
2	Bawang Putih	Bawang Merah Kecil	0.274989	0.238607	0.256798
3	Bawang Putih	Non-Woven Bag PcsM	0.334738	0.517686	0.426212
4	Chicken Leg Quarter	Non-Woven Bag PcsM	0.268559	0.443901	0.35623
5	Chicken Leg Quarter	Non-Woven Bag PcsS	0.222891	0.33069	0.27679
6	Ikan Kembong	Non-Woven Bag PcsM	0.123255	0.322983	0.223119
7	Sosis Borella 340g	Non-Woven Bag PcsM	0.25411	0.40496	0.329535
8	Sosis Perdix 340g	Non-Woven Bag PcsM	0.395725	0.578487	0.487106

The above table shows all of the important rules that are present in all of the 4 periods with their corresponding CVC, CVL, and OCVR values. a measurement of the variability in consumer purchasing habits is the overall variability of association rule analysis (OCVR). It is determined by averaging the support value variances of all association rules in a market basket. A high OCVR suggests extremely fluctuating client buying patterns, whereas a low OCVR shows generally steady customer buying patterns. Market baskets that are more likely to show variations in consumer purchasing behavior can be found using OCVR. Utilizing this knowledge, marketers can create marketing plans that are more successful in attracting certain market segments. For instance, a store may group items that are regularly bought together and close together to promote impulsive purchases.

OCVR can be used to monitor changes in consumers' purchasing trends over time. This data can be utilized to spot consumer behavior trends and create marketing plans that are better suited to capitalize on them.

To remove inventory and make room for new products, a merchant, for instance, could give discounts on items that are losing favors. Retailers and other companies that seek to comprehend

and affect consumer purchasing patterns should use OCVR as a helpful tool. Businesses can evaluate changes in customer buying patterns over time and identify market baskets that are more likely to show variability in consumer buying patterns by measuring OCVR. Using this knowledge, marketers can create marketing plans that are more successful at appealing to certain market patterns and baskets. Here is an illustration of how to create a marketing strategy with OCVR. Data on client transactions during the previous year was gathered by a store. The information indicates a high OCVR for the market basket of milk, bread, and eggs. This shows that consumer purchasing trends for these goods are quite erratic. This data can be used by the retailer to create marketing plans that will be more successful in attracting this market. The shop may, for instance, group these products together in the store, give these products discounts, or run advertisements that highlight these products[18]. A strong instrument that can be used to enhance marketing tactics is OCVR. Businesses can evaluate changes in customer buying patterns over time and identify market baskets that are more likely to show variability in consumer buying patterns by measuring OCVR. Using this knowledge, marketers can create marketing plans that are more successful at appealing to certain market patterns and baskets.

1.3. Design Marketing Strategy

Based on the insight gained from customer purchasing patterns and the association between items from the result of the market basket analysis, here are some marketing strategies that can be developed to optimize sales and improve customer satisfaction:

- **Find Popular Product Combinations**

  To find commonly occurring product combinations, examine the association rules produced by market basket analysis. These pairings show products that buyers regularly buy together.

- **The Configuration of the Shelves**

The supermarket can set up the shelves so that items that go well together are next to one another, such as placing an egg near cake or a green bean beverage near milk B.

- **Cross-Promotion and Bundling**

Using the observed popular product associations, create bundles of products or product combinations. Combine complementary goods and sell them as a set at a discount. This tactic can entice buyers to buy complementary goods and boost overall sales. Promote cross-selling by recommending relevant products to customers when they are making a purchase[19].

- **Promos that are Targeted**

Apply the association rules to create tailored promos that are directed at particular client groups. Send targeted offers or discounts on product B to consumers who have previously purchased product A, for instance, if customers who buy product A are likely to also buy product B. This tactic may help boost consumer loyalty and repeat business.

- **Visual Merchandising**

Base the design of your physical store or online shop on the relationships between the products. To promote cross-purchasing, group relevant items close together. To direct customers to relevant products, emphasize product suggestions or "Customers who bought this also bought" sections.

- **Inventory Control**

Modify your inventory control based on the relationships between the various products. To satisfy customer demand, make sure that frequently occurring products are always well stocked. Effective inventory management will reduce stockouts and increase sales possibilities[7].

- **Customer Segmentation**

  Divide your clientele into groups according to their spending habits and preferences. This might assist you in targeting particular client segments with your marketing messages and campaigns. You may design tailored marketing tactics to increase engagement and sales by having a thorough awareness of the particular requirements and preferences of various consumer groups.

Utilize the knowledge gained from the market basket study to pinpoint any areas where there are gaps or potential for new product development. Consider releasing a new product that complements or improves the current products if certain product associations are prevalent and buyers usually buy them together. Market basket analysis is a tool to gain insights into customer behavior, and the effectiveness of your marketing strategy will depend on the specific context and characteristics of your business. Continuously monitor and analyze customer data to refine your approach and adapt to changing customer preferences.

Chapter 5: Discussion and Future Works

5.1. Introduction

Market basket analysis using association rules has proven to be a valuable technique for understanding customer behavior, optimizing product offerings, and enhancing marketing strategies in the retail industry. However, there are several aspects that warrant further discussion and exploration. Firstly, the interpretation and actionable utilization of association rules require careful consideration. While association rules provide insights into item associations, it is important to distinguish between spurious correlations and meaningful relationships. Further research can focus on developing techniques to validate and prioritize association rules based on their relevance and impact on business outcomes.

Additionally, market basket analysis often relies on historical transaction data, which may not capture dynamic changes in customer preferences and market trends. Incorporating real-time or near-real-time data streams into market basket analysis can enable businesses to adapt their strategies in response to evolving customer behavior and market dynamics. Furthermore, market basket analysis has predominantly focused on transaction-level data, but there is an increasing need to incorporate contextual and demographic information to enhance the understanding of customer preferences and segmentation. Integrating external data sources, such as customer demographics, location, and sentiment analysis, can provide a more comprehensive view of customer behavior and enable personalized marketing strategies.

6. Discussion

the discussion and future works of market basket analysis using association rules revolve around advancing the techniques, incorporating contextual information, integrating machine learning, and leveraging big data technologies. By addressing these areas, businesses can unlock the full potential of market basket analysis and enhance their decision-making capabilities in the dynamic retail landscape. Integration with Other Data Sources: Market basket analysis can be enhanced by

integrating transactional data with other sources, such as customer surveys, social media data, and customer relationship management (CRM) systems. This integration can provide a more comprehensive understanding of customer preferences, attitudes, and behaviors, enabling businesses to refine their strategies and deliver more personalized experiences.

7. Future work

Advanced Association Rule Mining Techniques: Research efforts can be directed towards developing more sophisticated algorithms for association rule mining. This includes exploring techniques that handle high-dimensional and sparse datasets efficiently, address the scalability challenges associated with large-scale data, and consider additional constraints such as time and sequence dependencies.

- **Incorporating Machine Learning and Predictive Analytics**

Integrating machine learning algorithms and predictive analytics into market basket analysis can enhance its capabilities[20]. By leveraging techniques such as clustering, classification, and recommendation systems, businesses can go beyond association discovery and predict future purchasing patterns, enabling proactive decision-making and personalized customer experiences[21].

- **Contextual and Temporal Analysis**

Investigating the influence of context and time on association rules can provide deeper insights into customer behavior. This involves examining the impact of external factors, such as seasonality, promotions, and events, on item associations. By considering the temporal aspect, businesses can identify trends, patterns, and shifts in customer preferences over time[22].

- **Incorporating Big Data and Cloud Computing**

With the exponential growth of data in the digital era, leveraging big data technologies and cloud computing can enable scalable and efficient market basket analysis. Exploring distributed and parallel computing techniques can facilitate the analysis of large-scale datasets, allowing businesses to uncover meaningful associations and generate actionable insights in a timely manner[23].

Conclusion

Based on the completed analysis, it is possible to draw the conclusion that the results obtained in each time period were very varied and that they revealed distinct customer purchasing trends in each period. Eight rules with OCVR values less than 30% were found after additional examination of the rules using OCVR analysis. These rules can be used to create marketing strategies. To increase sales in Supermarket X, practice product bundling and shelf positioning. association mining rules are an effective method for market basket analysis, allowing companies to learn important details about consumer purchasing habits and item relationships. Businesses can optimize inventory management, boost product positioning, and improve marketing efforts by analyzing transactional data to determine which products are frequently purchased together. Businesses can better understand client preferences, cross-selling opportunities, and prospective product recommendations by developing and evaluating association rules.

These guidelines offer useful information that can be used to guide decisions, boost sales, and raise client satisfaction levels in general. Businesses can find patterns and associations that may not be immediately obvious by employing market basket analysis and association mining rules. Businesses can improve their capacity to target particular client segments, customize marketing campaigns, and optimize pricing strategies by utilizing these guidelines. Association mining rules also give companies the flexibility to adjust and react to shifting consumer preferences and market trends. Businesses may keep informed of changing consumer behavior and modify their plans by regularly analyzing transactional data and producing updated association rules. An effective methodology for organizations to obtain a competitive edge in the retail and e-commerce sectors is provided by association mining rules for market basket analysis. Businesses may increase income, increase customer satisfaction, and make data-driven decisions that lead to long-term success by comprehending client preferences, optimizing product offers, and customizing marketing strategies.

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Acknowledgements

I would like to express my sincere gratitude to everyone who contributed to the success of this project. Your assistance, guidance, and contributions have been really helpful in making it a success. Most of all I want to express my sincere gratitude to my supervisor Ms.张艳, for Her essential advice, knowledge, and constant encouragement during this process. The direction and quality of this work have been greatly influenced by her insightful criticism and mentorship.

I additionally want to express my gratitude to my friends, family, and colleagues for their support and compassion. Their support has given me the drive and inspiration I have needed to overcome challenges and complete this Thesis objectives.

Ruben Nar Ze Conis Silva Gusmão